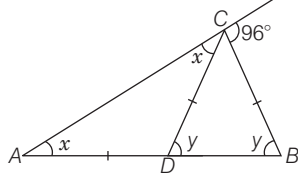


III. $\triangle PYX$ is similar to $\triangle PRQ$.

Hence, all statements are correct.

80. (c) See solved answer 17.

81. (b) Let $\angle CDB = y$ and $\angle CAD = x$
As, $AD = CD$



$$\Rightarrow \angle ACD = \angle CAD = x \text{ and } CD = CB$$

$$\Rightarrow \angle CBD = \angle CDB = y$$

From exterior angle property,

$$\angle CDB = \angle CAD + \angle ACD$$

$$\Rightarrow y = x + x \Rightarrow y = 2x$$

$$\text{Now, } \angle ACD + \angle DCB + 96^\circ = 180^\circ$$

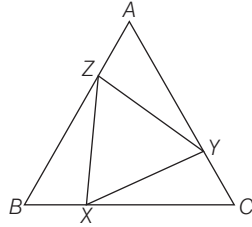
$$\Rightarrow x + 180^\circ - 2y + 96^\circ = 180^\circ$$

$$\Rightarrow \frac{y}{2} - 2y + 96^\circ = 0 \Rightarrow \frac{3}{2}y = 96^\circ$$

$$\therefore y = 64^\circ$$

82. (c) In an equilateral $\triangle ABC$,

$$AB = BC = CA \text{ and } \angle B = \angle C = \angle A$$



Given that, $BX = CY = AZ$

Now, in $\triangle XYC$, $\triangle ZYA$ and $\triangle XZB$,

$$BX = CY = AZ \quad \dots(i)$$

$$\Rightarrow (BC - XC) = (AC - AY)$$

$$= AB - BZ$$

$$\Rightarrow XC = AY = BZ \quad \dots(ii)$$

$$\text{and } \angle B = \angle C = \angle A \quad \dots(iii)$$

From Eqs. (i), (ii) and (iii), we get $\triangle XYC$, $\triangle ZYA$ and $\triangle XZB$ are congruent triangles.

$$\therefore XY = YZ = XZ$$

So, $\triangle XYZ$ is an equilateral triangle and $\triangle XYZ$ is similar to $\triangle ABC$.

So, both statements are correct.

83. (b) I. This statement is always true.

II. In any triangle three times the sum of squares of the sides of a triangle is equal to four times the sum of squares of medians

$$\text{i. e. } 3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)$$

$$\text{Similarly } 3(A'B'^2 + B'C'^2 + C'A'^2)$$

$$= 4(A'D'^2 + B'E'^2 + C'F'^2)$$

$$\therefore \frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{A'B'^2 + B'C'^2 + C'A'^2}{A'D'^2 + B'E'^2 + C'F'^2} = \frac{4}{3}$$

Hence, both statements are correct but statement II is not correct explanation of statement I.

$$\begin{aligned} 84. (c) \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} &= \frac{AB^2}{DE^2} \\ &= \frac{(\sqrt{2} + 1)^2}{(\sqrt{3})^2} = \frac{3 + 2\sqrt{2}}{3} \\ &= \frac{(3 + 2\sqrt{2})(3 - 2\sqrt{2})}{3(3 - 2\sqrt{2})} \\ &= \frac{1}{9 - 6\sqrt{2}} \end{aligned}$$

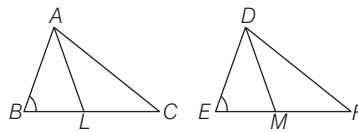
Hence, the required ratio is $1 : (9 - 6\sqrt{2})$.

85. (c) Given that, $\angle ABC = \angle DEF$,

$$\angle ACB = \angle DFE$$

and

$$\angle BAC = \angle EDF$$



So, $\triangle ABC$ and $\triangle DEF$ are similar.

$$\therefore \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

Now, L is the mid-point of BC , then

$$BL = \frac{1}{2}BC$$

Also, M is the mid-point of EF , then

$$EM = \frac{1}{2}EF \Rightarrow \frac{AB}{DE} = \frac{2BL}{2EM} = \frac{BL}{EM}$$

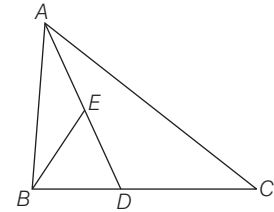
and $\angle ABL = \angle DEM$

$\therefore \triangle ABL$ is similar to $\triangle DEM$.

[By SAS Similarity]

Hence, statement I is true but statement II is false.

86. (c) In $\triangle ABC$, AD is the median and median AD bisects the area of $\triangle ABC$.
Area of $\triangle ABD = \frac{1}{2} \times \text{Area of } \triangle ABC$



Since, E is the mid-point of AD . Then, BE is the median of $\triangle ABD$. So, BE bisects area of $\triangle ABD$.

$$\begin{aligned} \text{I. Area of } \triangle BED &= \frac{1}{2} \times \text{Area of } \triangle ABD \\ &= \frac{1}{2} \left[\frac{1}{2} \times \text{Area of } \triangle ABC \right] \\ &= \frac{1}{4} \times \text{Area of } \triangle ABC \end{aligned}$$

Hence, statement I is correct.

II. Area of $\triangle ADC = \text{Area of } \triangle ABD$

$$= 2 \times \text{Area of } \triangle BED$$

So, statement II is also correct.

Hence, the statements I and II are correct.

24

QUADRILATERAL AND POLYGON

Regularly (2-3) questions have been asked from this chapter. Generally, questions from this chapter are tricky and mostly statement based. So, a clear concept of all the properties related to quadrilateral is necessary to do well.



QUADRILATERAL

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four angles and sum of these angles is equal to 360° .

Various types of quadrilateral are discussed below.

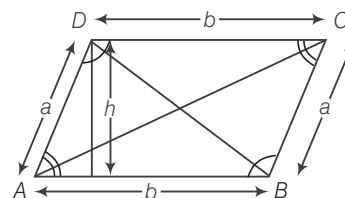
Parallelogram

A quadrilateral, in which opposite sides are parallel is called a parallelogram.

- (i) Area = Base $\times$ Height = $b \times h$ (ii) Perimeter = $2(a + b)$

Properties of Parallelogram

- (i) Diagonals of a parallelogram bisect each other.
- (ii) Each diagonal of a parallelogram divides it into two congruent triangles.
- (iii) Sum of any two adjacent angles is 180° .
- (iv) A parallelogram inscribed in a circle is a rectangle.
- (v) A parallelogram circumscribed about a circle is a rhombus.
- (vi) Two parallelograms have equal areas if they are on the same base and between the same parallel lines.
- (vii) Lines joining the midpoints of the sides of parallelogram is a parallelogram.
- (viii) The opposite angles of parallelogram are equal. ($\angle A = \angle C$ and $\angle B = \angle D$) (from above figure)
- (ix) The sum of the squares of the four sides is equal to the sum of squares of diagonal.
$$AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + AD^2$$



EXAMPLE 1. In a parallelogram $ABCD$, the bisectors of $\angle A$ and $\angle B$ meet at O . Then, the value of $\angle AOB$ is

- a. 55° b. 75° c. 90° d. 120°

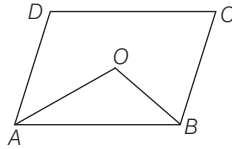
Sol. c. As, $ABCD$ is a parallelogram.

$\therefore \angle A + \angle B = 180^\circ$ [cointerior angle]

$$\frac{1}{2} \angle A + \frac{1}{2} \angle B = 90^\circ$$

$$\Rightarrow \angle OAB + \angle OBA = 90^\circ$$

$$\text{In } \triangle AOB, \angle AOB = 180^\circ - (\angle OAB + \angle OBA) = 180^\circ - 90^\circ = 90^\circ$$



EXAMPLE 2. In the adjoining figure, $ABCD$ is a parallelogram and E, F are the centroids of $\triangle ABD$ and $\triangle BCD$, respectively, then the length of EF is

- a. AE b. OB c. $\frac{1}{3} AE$ d. $\frac{1}{3} FC$

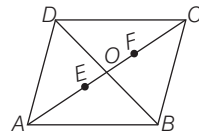
Sol. a. As E is the centroid of $\triangle ABD$ and AO is one of its medians.

$$\Rightarrow OA : EO = 3 : 1 \Rightarrow EO = \frac{1}{3} OA$$

$$\text{Similarly, } FO = \frac{1}{3} OC$$

$$\therefore EO + OF = \frac{1}{3} OA + \frac{1}{3} OC = \frac{1}{3} AC = AE$$

$$\therefore EF = AE$$

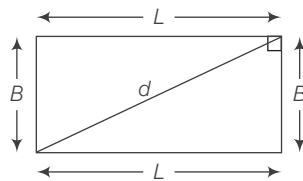


Rectangle

It is a parallelogram with opposite sides equal and each angle is equal to 90° .

(i) Area = Length $\times$ Breadth
 $= L \times B$

(ii) Perimeter = $2(L + B)$ (iii) Diagonal (d) = $\sqrt{L^2 + B^2}$

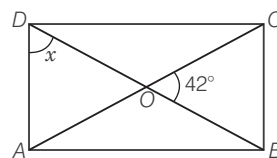


Properties of Rectangle

- The diagonals of a rectangle are of equal lengths and they bisect each other.
- All rectangles are parallelograms but reverse is not true.
- Diagonal of a rectangle inscribed in a circle is equal to the diameter of the circle.
- The figure formed by joining the mid-points of the sides of a rectangle is a rhombus.
- A rectangle and a parallelogram have equal area if they are on the same base and between same parallel lines.

EXAMPLE 3. In adjoining figure, if $ABCD$ is a rectangle and diagonals AC and BD intersect each other at O . Find the value of x

- a. 75° b. 70° c. 69° d. 63°



Sol. c. Since, diagonals of rectangle bisect each other

$$\therefore OD = OA \Rightarrow \angle ODA = \angle OAD$$

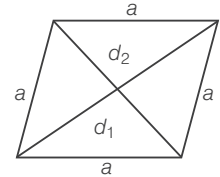
$$\text{Also, } \angle DOA = \angle COB = 42^\circ$$

$$\text{In } \triangle DOA, \angle DOA + \angle ODA + \angle OAD = 180^\circ$$

$$\Rightarrow 42^\circ + x + x = 180^\circ \Rightarrow 2x = 138^\circ \Rightarrow x = 69^\circ$$

Rhombus

It is a parallelogram with all 4 sides equal. The opposite angles in a rhombus are equal but they are not right angle.



(i) Area = $\frac{1}{2} \times d_1 \times d_2$

(ii) Perimeter = $4a$

(iii) Side (a) = $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$

(iv) $4a^2 = d_1^2 + d_2^2$

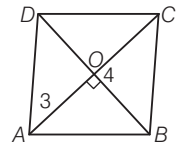
where, a = side, d_1 and d_2 are diagonals.

Properties of Rhombus

- The diagonals of a rhombus bisect each other at right angles (90°). But they are not necessarily equal.
- Diagonals bisect the vertex angles.
- The figure formed by joining the mid-points of the sides of rhombus is a rectangle.
- All rhombus are parallelogram but reverse is not true.
- A rhombus may or may not be a square but all squares are rhombus.

EXAMPLE 4. In a rhombus $ABCD$, diagonals intersect each other at O . If $AO = 3$ cm and $OB = 4$ cm, then find the perimeter of $ABCD$.

- a. 14 cm b. 30 cm c. 20 cm d. 28 cm



Sol. c. The diagonals of a rhombus are perpendicular bisectors. Which means they form right angles at the point of their intersection.

$\therefore$ In $\triangle AOB$, by pythagoras theorem,

$$AO^2 + OB^2 = AB^2 \Rightarrow AB = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}$$

$$\therefore AB = 5 \text{ cm}$$

Hence, perimeter of rhombus

$$ABCD = AB + BC + CD + DA = 5 + 5 + 5 + 5 = 20 \text{ cm}$$

Square

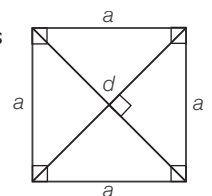
It is a parallelogram with all four sides equal and each angle is equal to 90° .

(i) Area = (side) 2 = a^2 or $\frac{1}{2} d^2$

(ii) Perimeter = $4 \times \text{side} = 4a$

(iii) Diagonal (d) = $a\sqrt{2}$

where, a = side, d = diagonal

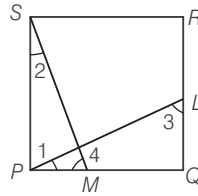


Properties of Square

- (i) Diagonal of a square are equal and bisect each other at right angles (90°).
- (ii) All square are rhombus but converse is not true.
- (iii) Diagonal of square is the diameter of the circumscribing circle that circumscribes the square.
- (iv) Side of a circumscribed square is equal to the diameter of the inscribed circle.
- (v) The figure formed by joining the mid-points of the sides of square is a square.

EXAMPLE 5. PQRS is a square, P is joined to a point L on QR and S is joined to a point M on PQ. If $PL = SM$, then

- a. $\angle POM > 90^\circ$
- b. $\angle POM = 90^\circ$
- c. $\angle POM < 90^\circ$
- d. None of these

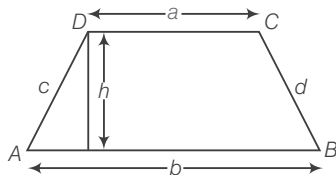


Sol. b. In $\triangle SPM$ and $\triangle PQL$, $\angle SPM = \angle PQL = 90^\circ$

$$\begin{aligned}
 &SP = PQ \text{ and } SM = PL \Rightarrow \triangle SPM \cong \triangle PQL \quad [\text{By AAS}] \\
 \Rightarrow &\angle 1 = \angle 2 \text{ and } \angle 3 = \angle 4 \\
 \text{In } \triangle SPM, &\quad \angle 2 + 90^\circ + \angle 4 = 180^\circ \Rightarrow \angle 2 + \angle 4 = 90^\circ \\
 \text{In } \triangle OPM, &\quad \angle 1 + \angle 4 + \angle POM = 180^\circ \\
 \Rightarrow &\quad \angle 2 + \angle 4 + \angle POM = 180^\circ \\
 \Rightarrow &\quad 90^\circ + \angle POM = 180^\circ \quad [\because \angle 2 + \angle 4 = 90^\circ] \\
 \Rightarrow &\quad \angle POM = 90^\circ
 \end{aligned}$$

Trapezium

It is a quadrilateral with any one pair of opposite sides parallel.

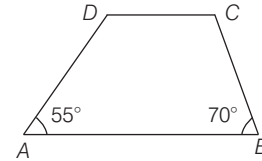


- (i) Area = $\frac{1}{2}$ (Sum of the parallel sides $\times$ Height)
 $= \frac{1}{2} (a + b)h$ where, a and b are parallel sides and h is the height or perpendicular distance between a and b .
- (ii) Perimeter = $AB + BC + CD + AD$

Properties

- (i) If the non-parallel sides are equal then the diagonals will also be equal to each other.
- (ii) Diagonals intersect each other proportionally in the ratio of lengths of parallel sides.
- (iii) If a trapezium is inscribed in a circle, then it is an isosceles trapezium with equal oblique sides.
- (iv) The line joining the mid-points of non-parallel sides is half the sum of parallel side and is called median.
i.e. Median = $\frac{1}{2} \times$ sum of parallel sides.

EXAMPLE 6. In the adjoining figure, ABCD is a trapezium in which $AB \parallel DC$. If $\angle A = 55^\circ$ and $\angle B = 70^\circ$, then the value of $\angle C$ and $\angle D$ is



- a. 75° and 85°
- b. 90° and 120°
- c. 110° and 125°
- d. 115° and 120°

Sol. c. As, $AB \parallel CD$

$$\text{So, } \angle A + \angle D = 180^\circ \Rightarrow \angle D = 180^\circ - 55^\circ = 125^\circ$$

$$\text{Also, } \angle B + \angle C = 180^\circ \Rightarrow \angle C = 180^\circ - \angle B$$

$$\Rightarrow \angle C = 180^\circ - 70^\circ = 110^\circ$$

$$\text{So, } \angle C = 110^\circ \text{ and } \angle D = 125^\circ$$

POLYGON

A polygon is a closed, plane figure bounded by 'n' straight lines ($n \geq 3$). Each of the line segment forming the polygon is called its sides.

A polygon may be a triangle, quadrilateral, pentagon etc. Polygons are classified according to the number of sides as given below:

Number of sides	Name
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

Regular Polygon

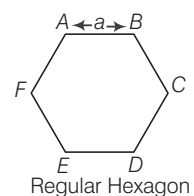
In regular polygons, all sides and all interior angles are equal. If each side of a regular polygon of n sides = a , then

$$(i) \text{ Area of regular pentagon} = 5a^2 \frac{\sqrt{3}}{4}$$

$$(ii) \text{ Area of regular hexagon} = 6a^2 \frac{\sqrt{3}}{4}$$

$$(iii) \text{ Area of regular octagon} = 2(\sqrt{2} + 1)a^2$$

$$(iv) \text{ Each exterior angle} = \frac{360^\circ}{n}$$



Regular Hexagon

(v) Each interior angle $= 180^\circ - \text{Exterior angle}$

$$= \frac{(n-2) \times 180}{n}$$

(vi) Number of diagonals $= \left\{ \frac{n(n-1)}{2} - n \right\}$

(vii) Sum of all interior angles $= (n-2) \times 180^\circ$.

EXAMPLE 7. A polygon has 35 diagonals. Then, the number of sides of that polygon is

- a. 7 b. 10 c. 11 d. 12

Sol. b. Let number of sides be n , then

$$\begin{aligned} \frac{n(n-1)}{2} - n &= 35 \Rightarrow \frac{n^2 - n - 2n}{2} = 35 \\ \Rightarrow n^2 - 3n - 70 &= 0 \Rightarrow n^2 - 10n + 7n - 70 = 0 \\ \Rightarrow n(n-10) + 7(n-10) &= 0 \Rightarrow (n-10)(n+7) = 0 \\ \Rightarrow n &= 10 \text{ and } n \neq -7 \end{aligned}$$

Hence, the number of sides of the polygon is 10.

EXAMPLE 8. The angles of a hexagon are x° , $(x-5)^\circ$, $(x-5)^\circ$, $(2x-5)^\circ$, $(2x-5)^\circ$ and $(2x+20)^\circ$. Then, the value of x is

- a. 60° b. 75° c. 80° d. 90°

Sol. c. Sum of interior angles of a hexagon $= (n-2) \times 180^\circ$

$$= 4 \times 180^\circ = 720^\circ$$

$$\begin{aligned} \therefore x^\circ + (x-5)^\circ + (x-5)^\circ + (2x-5)^\circ \\ + (2x-5)^\circ + (2x+20)^\circ &= 720^\circ \\ \therefore 9x &= 720^\circ = \frac{720^\circ}{9} = 80^\circ \end{aligned}$$

Hence, the value of x is 80° .

EXAMPLE 9. The difference between the interior and exterior angles of a regular polygon is 60° . Then, how many sides are there in that polygon?

- a. 5 b. 6 c. 7 d. 8

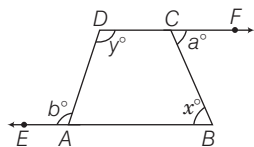
Sol. b. Here, (interior angle) $-$ (exterior angle) $= 60^\circ$

$$\begin{aligned} \Rightarrow \frac{(n-2) \times 180}{n} - \frac{360}{n} &= 60 \\ \Rightarrow \frac{1}{n} [(n-2) \times 180 - 360] &= 60 \\ \Rightarrow \frac{1}{n} [180n - 360 - 360] &= 60 \Rightarrow \frac{1}{n} [180n - 720] = 60 \\ \Rightarrow 180n - 720 &= 60n \Rightarrow 180n - 60n = 720 \\ \Rightarrow 120n &= 720 \Rightarrow n = 6 \end{aligned}$$

Therefore, the polygon contains 6 sides.

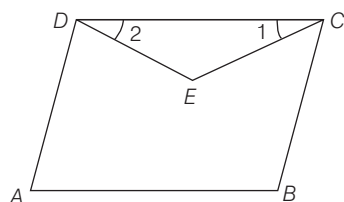
> PRACTICE EXERCISE

1. The sides BA and DC of quadrilateral $ABCD$ are produced as shown in figure. Then, which of the following statement is correct?



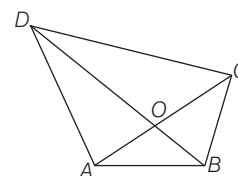
- (a) $2x^\circ + y^\circ = a^\circ + b^\circ$ (b) $x^\circ + \frac{1}{2}y^\circ = \frac{a^\circ + b^\circ}{2}$
(c) $x^\circ + y^\circ = a^\circ + b^\circ$ (d) $x^\circ + a^\circ = y^\circ + b^\circ$

2. In the quadrilateral $ABCD$, the line segments bisecting $\angle C$ and $\angle D$ meet at E . Then, the correct statement is



- (a) $\angle A + \angle B = \angle CED$ (b) $\angle A + \angle B = 2 \angle CED$
(c) $\angle A + \angle B = 3 \angle CED$ (d) None of these

3. If $ABCD$ is a quadrilateral whose diagonals AC and BD intersect at O , then



- (a) $(AB + BC + CD + DA) < (AC + BD)$
(b) $(AB + BC + CD + DA) > 2(AC + BD)$
(c) $(AB + BC + CD + DA) > (AC + BD)$
(d) $(AB + BC + CD + DA) = 2(AC + BD)$

4. If area of a parallelogram with sides p and q is R and that of a rectangle with sides p and q is S , then

- (a) $R > S$ (b) $R < S$
(c) $R = S$ (d) None of these

5. Two parallelogram stand on equal bases and between the same parallel. The ratio of their areas is

- (a) 1:2 (b) 2:1 (c) 1:3 (d) 1:1